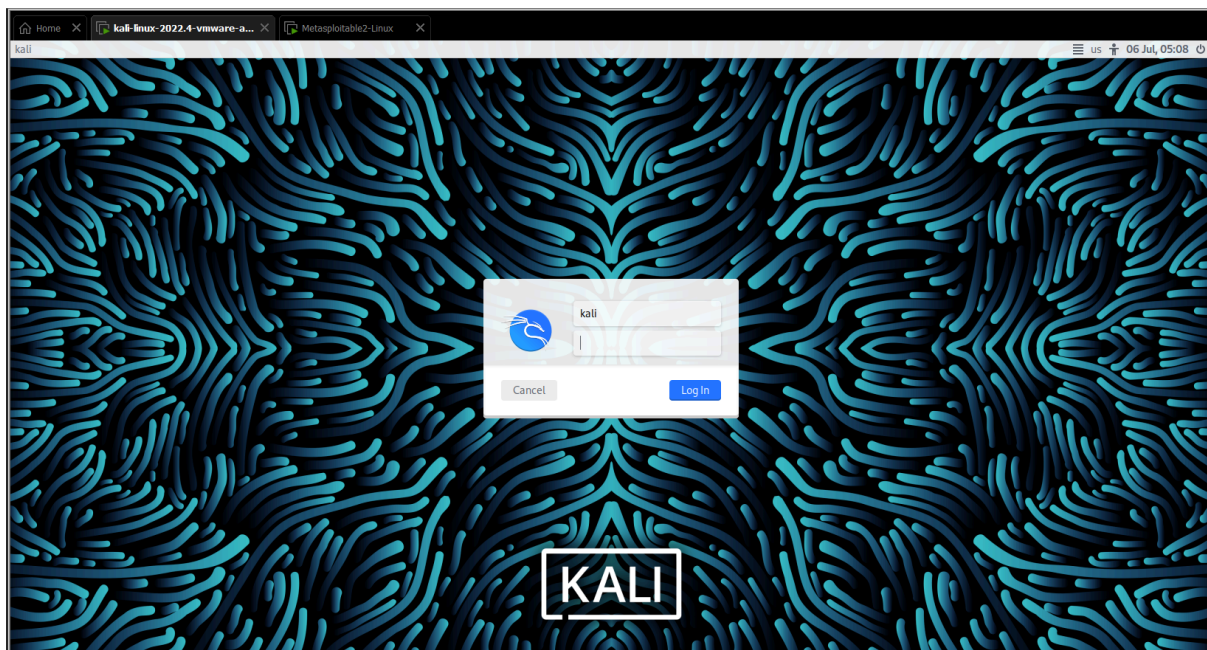
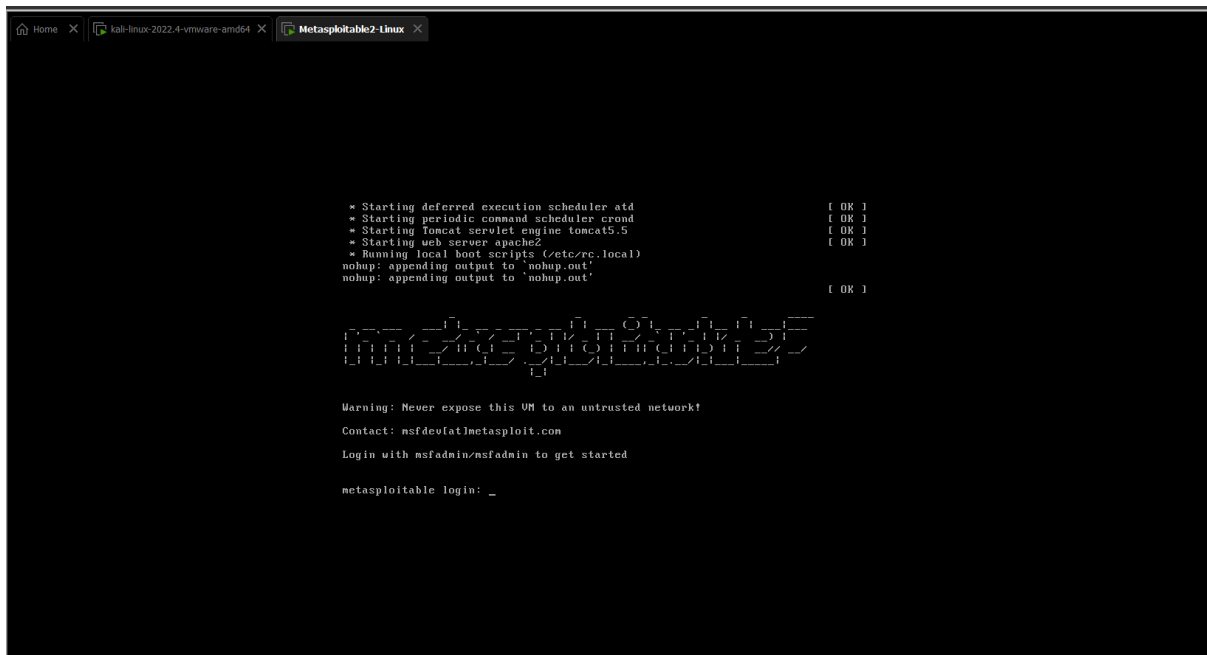


Exploiting SSH Vulnerabilities in Metasploitable 2

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This documentation explores the process of exploiting SSH vulnerabilities within the Metasploitable 2 environment. Metasploitable 2, a deliberately vulnerable virtual machine, serves as the testing ground for understanding and demonstrating the exploitation of SSH weaknesses.

Open both machines, Kali Linux and Metasploitable.



Run the below command in Metasploitable 2 to find the IP Address.

>>>Ifconfig

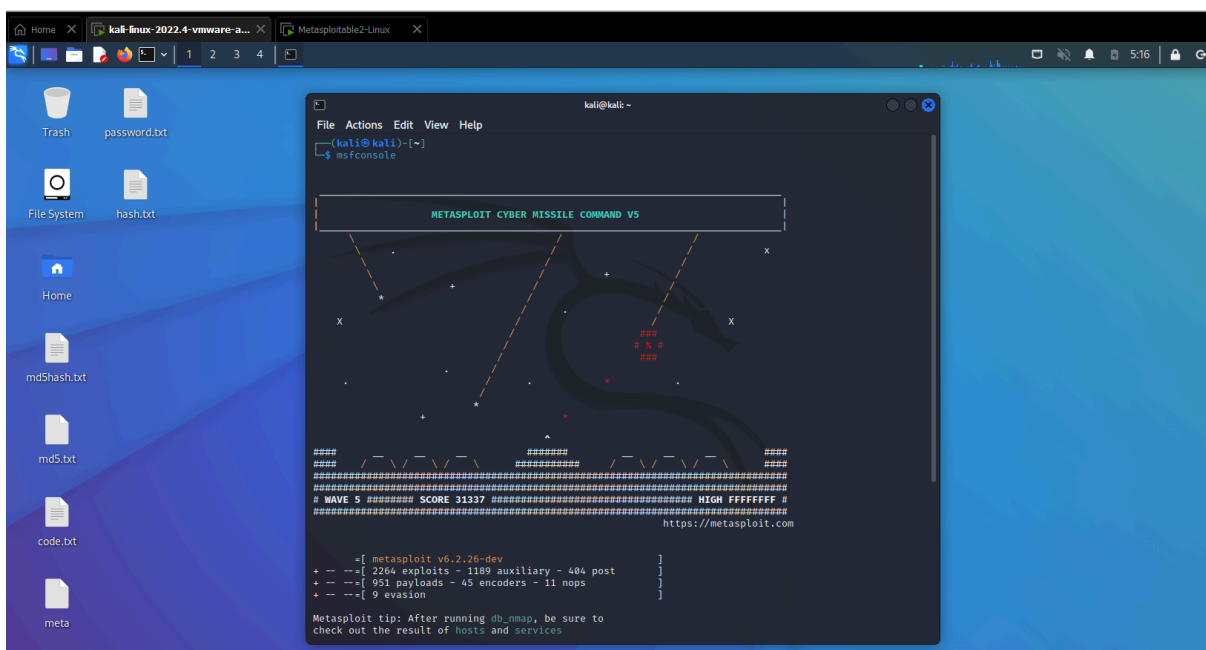
```
Home X kali-linux-2022.4-vmware-amd64 X Metasploitable2-Linux X

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:2f:87:48
          inet addr:192.168.107.128  Bcast:192.168.107.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe2f:8748/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:342 errors:0 dropped:0 overruns:0 frame:0
          TX packets:87 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:22777 (22.2 KB)  TX bytes:12207 (11.9 KB)
          Interrupt:17 Base address:0x2000

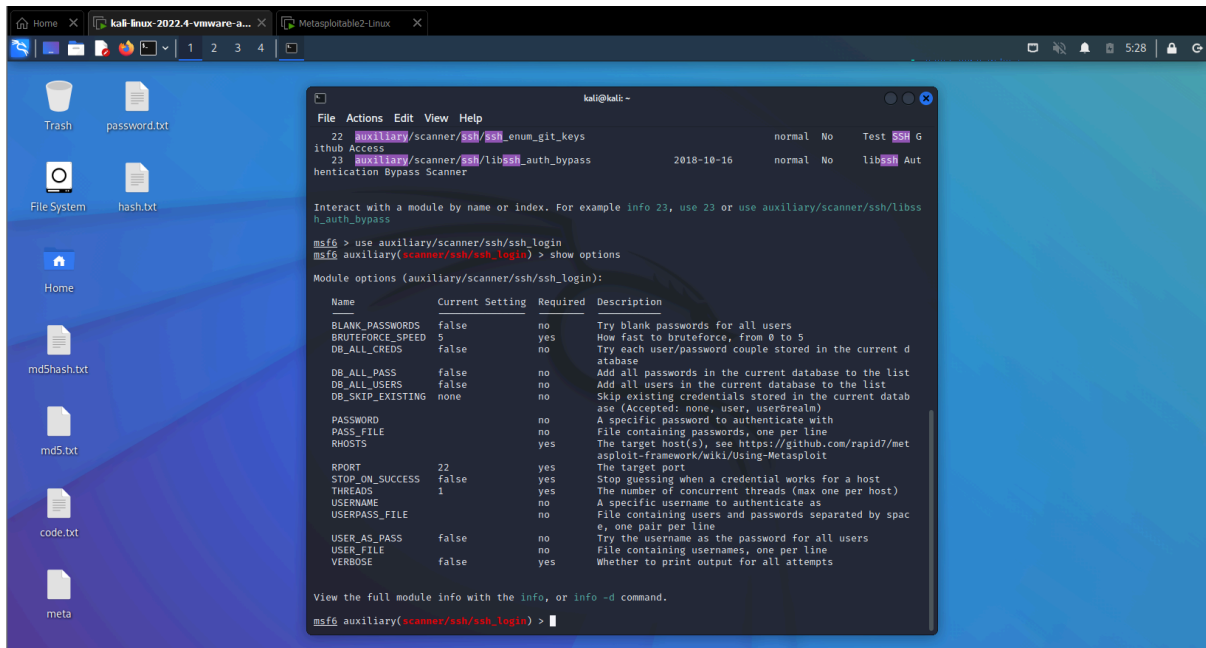
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:210 errors:0 dropped:0 overruns:0 frame:0
          TX packets:210 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:81265 (79.3 KB)  TX bytes:81265 (79.3 KB)

msfadmin@metasploitable:~$ _
```

Now, open Kali Linux and open Metasploit framework using the **msfconsole** command.



Now, we are to search for ssh_login Auxiliaries by using the **Search auxiliary ssh** command in msfconsole.



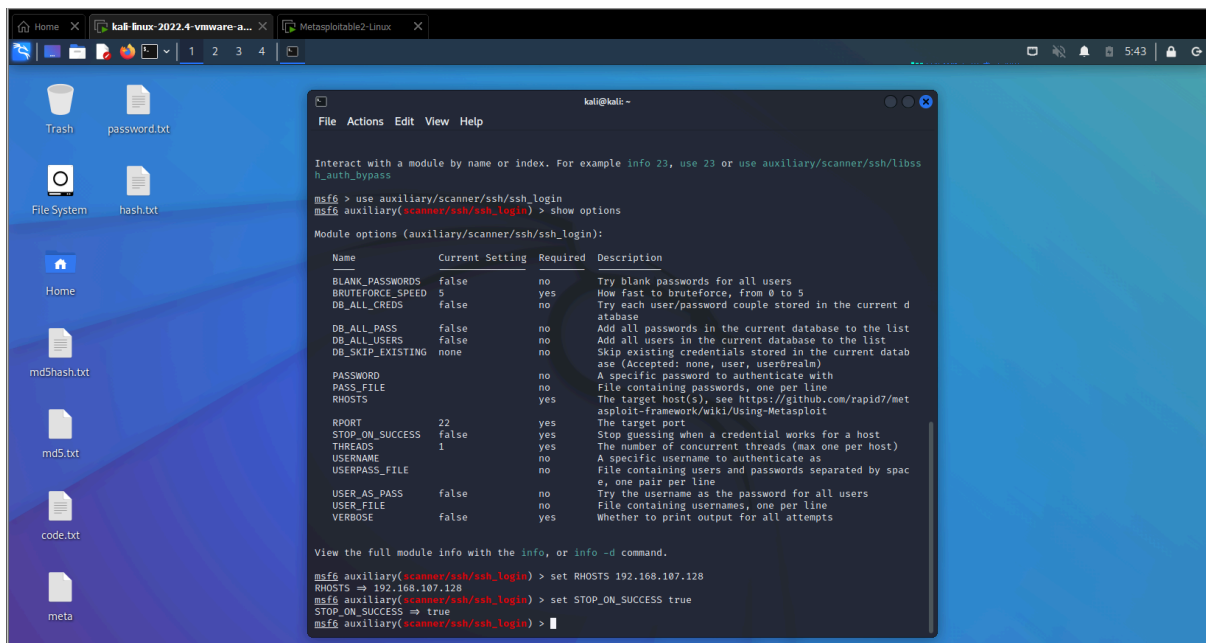
Now, we can see that we have a bunch of different options to set before launching our attack. Now set the required option and launch the attack.

Set the options that are required with the set command as followed in the image below

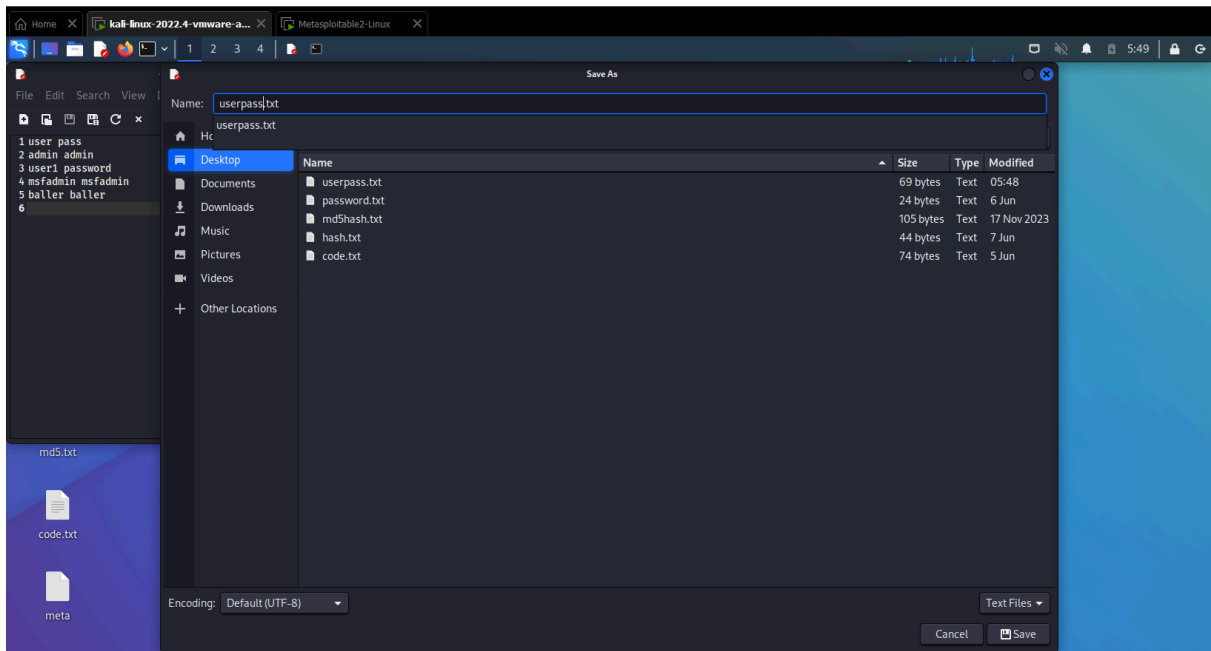
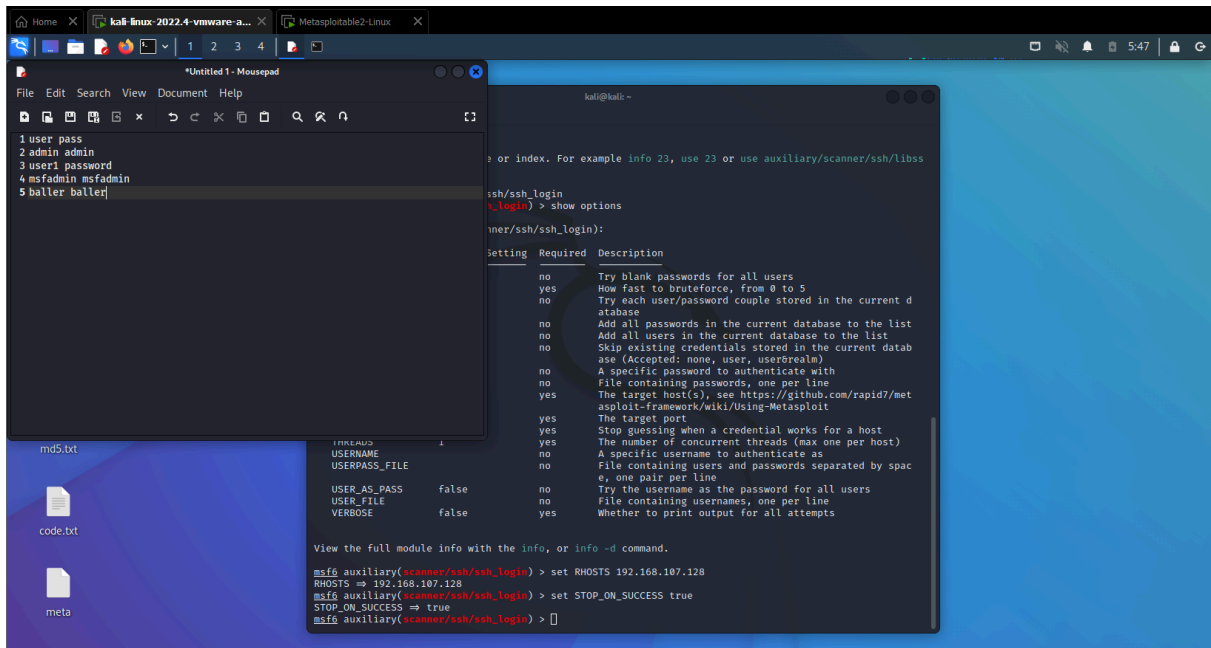
>>>set RHOST 192.168.107.128 (The target host(s))

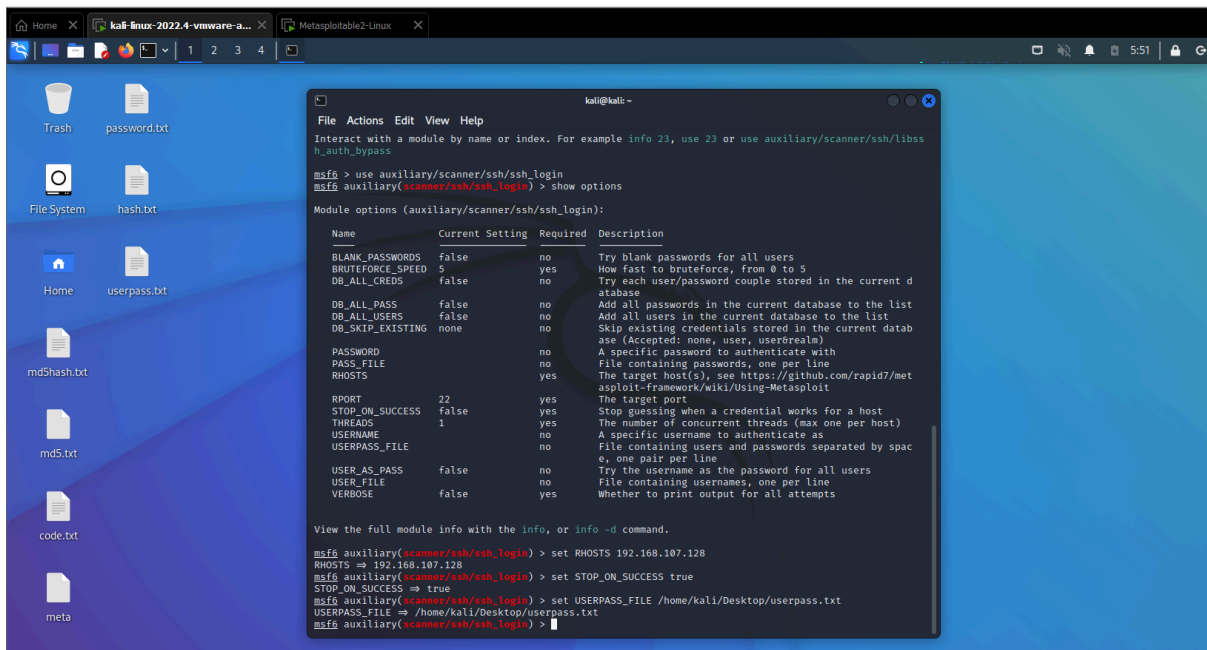
>>>set STOP_ON_SUCCESS true (Stop guessing when a credential works for a host)

>>>set USERPASS_FILE /path/to/userpass.txt (File containing users and passwords separated by space, one pair per line)

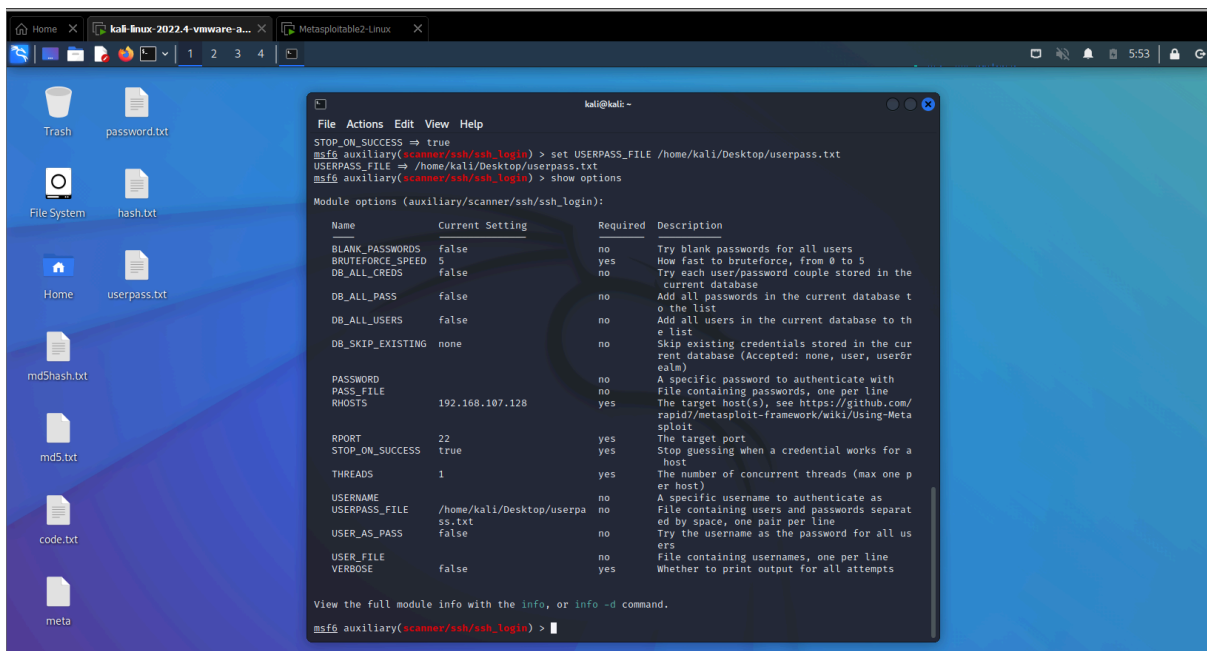


Now, i will create a USERPASS_FILE txt that contains both usernames and passwords, separated by a space.

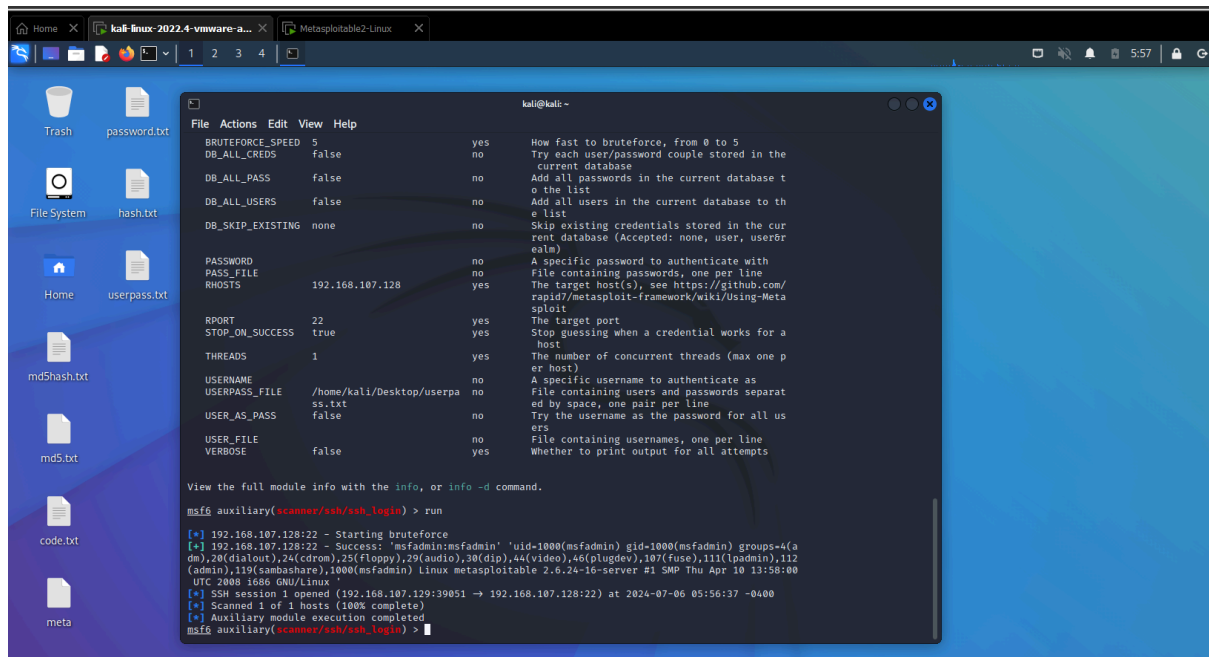




Now, let's type the **show options** commands to see the changes we have made.



We are all set to go and now we can launch the attack and watch each attempt on the terminal. To launch the attack, we will use the **run** command.



After inputting the **run** command, it brute forced into the system and when the attack was successful, it returned the password and username, which is msfadmin msfadmin.

Conclusion

The project on exploiting SSH vulnerabilities in Metasploitable 2 provides a comprehensive exploration of how security weaknesses in SSH can be identified and leveraged using various tools and techniques. Through this project, several critical insights and conclusions have been drawn:

1. **Understanding Metasploitable 2:** Metasploitable 2, as a vulnerable Linux machine, serves as an excellent platform for learning and practising penetration testing. The intentional vulnerabilities present in this environment mimic real-world scenarios, enabling ethical hackers to hone their skills.
2. **SSH Vulnerabilities:** Secure Shell (SSH) is a crucial protocol for secure remote login and other secure network services. However, misconfigurations and outdated software can expose SSH to various vulnerabilities. This project identified several common SSH vulnerabilities, including weak passwords, outdated SSH versions, and improper key management.
3. **Exploitation Techniques:**
 - o **Version-Specific Exploits:** Older versions of SSH servers may contain known vulnerabilities that can be exploited. For instance, the use of Metasploit Framework allowed the exploitation of specific vulnerabilities associated with the outdated SSH version running on Metasploitable 2.

- **Credential Harvesting:** Exploiting SSH misconfigurations can lead to unauthorised access and credential harvesting, further enabling lateral movement within a network.
- 4. **Metasploit Framework:** The Metasploit Framework proved to be an invaluable tool in this project. Its extensive library of exploits, payloads, and auxiliary modules facilitated the exploitation process, demonstrating the potential damage an attacker could inflict if these vulnerabilities were present in a production environment.
- 5. **Mitigation Strategies:** The project emphasised the importance of several mitigation strategies to secure SSH:
 - **Regular Updates:** Keeping SSH software up-to-date to protect against known vulnerabilities.
 - **Strong Authentication:** Implementing strong, unique passwords and considering multi-factor authentication (MFA).
 - **Proper Key Management:** Ensuring proper generation, storage, and rotation of SSH keys.
 - **Configuration Hardening:** Disabling root login, using non-standard ports, and applying strict access controls.
- 6. **Ethical Considerations:** The project reinforced the ethical responsibilities of cybersecurity professionals. While the techniques demonstrated can be powerful, they must be used within legal and ethical boundaries, focusing on improving security rather than causing harm.

Final Thoughts

This project not only highlighted the potential risks associated with SSH vulnerabilities but also demonstrated the critical need for ongoing vigilance in cybersecurity practices. As technology evolves, so do the methods and tools available to attackers. Therefore, continuous education, regular security assessments, and proactive measures are essential to safeguarding systems against exploitation.

The insights gained from exploiting SSH vulnerabilities in Metasploitable 2 underscore the importance of a robust security posture. By understanding and addressing these vulnerabilities, organisations can better protect their assets, maintain trust, and ensure the integrity of their systems.